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Identifying soybean lines differing in gas exchange sensitivity to humidity

BY JAMES A. BUNCE

Light and Plant Growth Laboratory, U.S.D.A.-A.R.S. Beltsville, Maryland 20705, U.S.A.

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SUMMARY

The ratios of root length and root weight to leaf area differed within and between cultivars of soybean. Plants with low ratios of root length or weight to leaf area had leaf conductances and net photosynthetic rates more reduced by a given increase in the leaf to air water vapour pressure difference around a single leaf than plants with high ratios. Plant and root system conductances to water were estimated as transpiration rate per unit leaf area divided by the difference between substrate and leaf water potentials, and by the rate of water flow through pressurised root systems. These conductances were greater in plants with large, as compared with small, root systems per unit leaf area. Cultivar rankings in sensitivity of gas exchange to humidity were consistent in controlled environment chambers and in field tests.

INTRODUCTION

It was previously noted that the water vapour content of the air was often low enough to considerably reduce net photosynthesis and stomatal conductance of upper canopy leaves of soybeans under field conditions in Beltsville, Maryland (Bunce, 1982). I have attempted to identify locally grown soybean lines with different sensitivities of net photosynthesis and stomatal conductance to atmospheric water content. Such lines should facilitate the testing of the possible benefits of altered sensitivity. The sensitivity of stomatal conductance to the leaf:air water vapour pressure difference (VPD) was correlated with the ratio of total root length to total leaf area in previous comparisons between species and between growth environments (Bunce, 1981). Those results suggested that the ratio of root length to leaf area might be useful in selecting plants within a species which differ in the sensitivity of leaf conductance to VPD. Because VPD may have a variable effect on net photosynthesis relative to its effect on leaf conductance (Bunce, 1984), ranking of plants in sensitivity of leaf conductance might not be the same as ranking in sensitivity of net photosynthesis. In this paper are reported results of attempts to use a morphological criterion to identify lines within the species *Glycine max* L. Merr. with different sensitivities of net photosynthesis and leaf conductance to VPD. Rankings of sensitivity determined in controlled environments were compared with rankings under field conditions.

METHODS AND MATERIALS

Identification of lines. Plants were characterised initially by the ratio of root dry weight to plant leaf area. Root weight was used rather than root length since it could be measured much more quickly. Differences in the ratio of root length to leaf area were tested later. The method of Newman (1966) was used to measure the length of the total root system.

Eight plants each of the cultivars Kent, Williams, Clark, Dorman, and Lincoln of *Glycine max* were grown one plant per pot in vermiculite in 15 cm diameter plastic pots and were

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flushed daily with a complete nutrient solution (Silvius, Kremer & Lee, 1978). Air temperature was 28°C in the light and 22°C in the dark. Dewpoint was 18°C. There were 14 h of light per day at 1000 μmol m^{-2} s^{-1} photosynthetic photon flux density (PPFD) from a mixture of high pressure sodium and multivapour lamps. Plant leaf areas and root dry weights were measured 17 days after planting, when leaf areas were about 2·5 dm^2 per plant.

Selfed progeny of 10 plants of the cv. Kent were compared for the ratio of root weight to leaf area. Seven seeds from each parent were grown as described above. Plants were harvested 21 days after planting, when leaf areas were approximately 5 dm^2 per plant. Two lines were found to differ significantly in the ratio of root weight to leaf area and were used in subsequent tests.

Characterisation of gas exchange sensitivity. Measurements were made both in controlled environment chambers and outdoors.

For the controlled environment work, cvs Kent, Clark, and Dorman, and two lines within Kent were grown at 28/22°C air temperature, 18°C dewpoint, with 14 h light per day at 1000 μmol m^{-2} s^{-1} PPFD. Plants were grown in vermiculite and pots were flushed each day with nutrient solution. Gas exchange was measured when total leaf areas were between 5 and 10 dm^2. Steady state water vapour and carbon dioxide exchange rates were determined at VPD s of 1·0 and 3·0 kPa on the terminal leaflets of either the second or third trifoliolate leaf from the bottom (standardised within specific comparison). In either case, the measured leaves were the most recent fully expanded mainstem leaves. Measurements were made in an open gas exchange system in which leaf conductance to water vapour and net carbon dioxide uptake rates were measured at constant ambient carbon dioxide concentration (345 cm^3 m^{-3}), PPFD (1500 μmol m^{-2} s^{-1}), leaf temperature (30 or 33°C), with various dewpoints. Air speed was 2 m s^{-1}. Leaf dimensions were similar for all lines, and no separation of boundary layer and leaf conductances to water vapour was made. Boundary layer conductances were about five times the maximum total conductances. The rest of the plant was kept either in the growth chamber, or else at 30°C air temperature, 1500 μmol m^{-2} s^{-1} PPFD, 1·0 kPa VPD while the response of the single leaflet was determined. Plants were in the measurement environment for 2 h before measurements were begun. Leaves were measured first at low VPD, after exposure to low VPD for 2 h. VPD was then increased and gas exchange again measured after being stable at the new VPD for 1 h.

Field experiments were conducted for two growing seasons at Beltsville using cvs Kent, Clark and Dorman. Measurements were made several times during the seasons and plants were older than any measured in the controlled environment experiments. Plants had not experienced significant soil drying in either season. In one season gas exchange measurements were made near midday on clear days using the same open gas exchange system used in the controlled environment studies. Terminal leaflets of fully expanded upper canopy leaves were measured at 33°C, and 2000 μmol m^{-2} s^{-1} PPFD, and 1·5 kPa VPD, and also at the ambient dewpoint (giving a mean VPD of 3·0 kPa). The rest of the shoot was at the ambient air temperature, dewpoint and PPFD. Xylem pressure potential measurements were made using a pressure chamber on five leaves per cultivar at each measurement time. In the other growing season, measurements of leaf conductance to water vapour were made with a transit time diffusion porometer. Measurements were made when PPFD exceeded 1500 μmol m^{-2} s^{-1}. Upper and lower leaf surface conductances were measured on 6 to 10 leaves of each cultivar at each time. Upper and lower surface conductances were summed. Days were chosen to provide a range of VPD's. Leaf water potentials of three excised discs were measured *simultaneously* on each cultivar with a dewpoint hygrometer each measurement day.

Plant and root conductance. An estimate of plant conductance to water was made from measurement of transpiration rate and leaf water potential for cvs Clark, Kent, and Dorman, and the two lines of Kent. Plants were grown in vermiculite at 28/22°C air temperature, 18°C dewpoint, 1000 μmol m^{-2} s^{-1} PPFD for 14 h per day until leaf areas were about 5 dm^2.

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Plants were then placed in a gas exchange chamber with air temperature 30°C, dewpoint 10°C, and 1800 μmol m^{-2} s^{-1} PPFD for 3 h. Transpiration rate was monitored in an open gas exchange system, and measurements were made of water potentials of leaf discs excised at the end of the 3 h. Conductance was estimated as the rate of transpiration divided by the difference between the osmotic potential of the nutrient solution (−0·08 MPa) and the leaf water potential. Transpiration was expressed per unit of leaf area, so that conductances are also per unit of leaf area.

Root system conductance was estimated by measuring the rate of water flow through pressurised root systems. Plants were grown in vermiculite in the same way as for the plant conductance measurements until they had about 2 dm^2 leaf area. The stem was severed above the cotyledonary node, and the pot was placed in a pressure chamber with the cut end of the stem outside. The air around the pot was pressurised with N_2 to 1·0 MPa. Liquid emerging from the cut end of the stem was collected and weighed at 5-min intervals until a consistent rate of flow was achieved (about 20 min after pressurisation). The flow rate remained constant for more than an hour after the initial 20-min period. A slow equilibration of water flow rates through root systems has been previously reported (Kuiper & Kuiper, 1974). Leaf area and root weight were determined.

Statistics. Analysis of variance was used to calculate the L.S.D. at the 0·05 level, in cases where the F test was significant.

RESULTS

In all cases in which carbon dioxide uptake rate was measured simultaneously with leaf conductance to water vapour, the two changed proportionally in response to VPD. A 50% decrease in leaf conductance was associated with a 20% decrease in the rate of net carbon dioxide fixation. Because of this proportionality, the ranking of lines in sensitivity of net photosynthesis to VPD was the same as the ranking in leaf conductance response.

The five soybean cultivars had root weight to leaf area ratios ranging from 14·3 to 21·2 g m^{-2} (Table 1) as seedlings. Further work was done only with cvs Clark, Kent, and Dorman, representing high, intermediate, and low ratios, respectively. Measurements on larger plants grown in a controlled environment showed that there were differences in the ratio of root system length to leaf area, and in the reduction in leaf conductance at high VPD (Table 1). Field measurements of leaf conductance at high and low VPD at controlled PPFD and temperature also showed that conductance was most sensitive to VPD in Dorman, and least sensitive in Clark (Table 2). Leaf conductances at 1·5 kPa VPD were not significantly different between cultivars. Xylem pressure potentials averaged −1·20 MPa, with no

Table 1. *Ratios of root length and weight to leaf area, and leaf conductance (c_s) to water vapour at 3·0 compared to 1·0 kPa leaf to air water vapour pressure difference for five cultivars of* Glycine max. *Ratios of root weight to leaf area are for plants with about 2·5 dm^2 leaf area; those of root length to leaf area are for plants with 5 to 10 dm^2 leaf area. Conductances were measured at 33° C leaf temperature under controlled environment conditions*

Cultivar	Root wt/area (g m^{-2})	Root length/area (m m^{-2})	c_s at 3·0/ c_s at 1·0
Clark	21·2	1470	0·76
Dorman	14·3	1070	0·58
Kent	16·5	1200	0·64
Lincoln	19·3	—	—
Williams	17·5	—	—
L.S.D.	2·0	75	0·04

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significant differences between cultivars. Measurements with a diffusion porometer under uncontrolled conditions in the field also showed the same pattern in cultivar sensitivity to VPD (Table 2). Leaf water potentials averaged −0·97 MPa at low VPDs and −1·23 MPa at high VPDs, with no significant differences between cultivars.

The two lines from within cv. Kent differed in the ratio of root weight to leaf area, and in the sensitivity of leaf conductance to VPD under controlled environment conditions (Table 3). Line A had a ratio of root length to leaf area of 1690 m m^{-2}, and line B a ratio of 1420 m m^{-2}. Line A was less sensitive to VPD than the unselected Kent, and line B was more sensitive.

Plant and root conductances were highest in Clark and lowest in Dorman, showing the same pattern as root weight per leaf area and the sensitivity of leaf conductance to VPD (Table 3).

DISCUSSION

The results showed that there were differences between soybean cultivars in the sensitivity of net photosynthesis and leaf conductance to VPD and in the ratios of root length and weight to leaf area. Plants with low ratios had greater sensitivity of both net photosynthesis and leaf conductance to VPD. Cultivars differed in the sensitivity to VPD under controlled environment conditions and also in two different field comparisons. The ranking in sensitivity was the same in all tests. The results also showed that there was sufficient diversity within a cultivar to identify seedlots with significant differences in sensitivity of gas exchange to VPD. In both of these situations, the morphological criterion was capable of identifying lines with significant differences in physiological response. Differences in plant and root conductances were found in the lines differing in the ratio of root length to leaf area, confirming that root system size correlates with conductance, at least in these intraspecific comparisons.

These experiments were not designed to test the causality of the correlation between the sensitivity of leaf conductance to VPD and root conductance. However, the field observations at high VPD suggest that leaf conductances were adjusted to VPD such that leaf water

Table 2. *Leaf conductance to water vapour under field conditions for three cultivars of* Glycine max. *Expt 1: measurements were made with an open gas exchange system; leaf temperature 33° C, photosynthetic photon flux density (PPFD) 2000 μmol m^{-2} s^{-1}. Conductances (c_s) at ambient dewpoint (average leaf to air vapour pressure difference (VPD) 3·0 kPa were compared with conductances at 1·5 kPa. Expt 2: measurements were made with a diffusion porometer on days with high or low VPD, PPFD 1500 μmol^{-2} s^{-1}*

Experiment	Cultivar	Conductance c_s at 3·0/ c_s at 1·5	Number of observations (number of leaves)
1	Clark	0·85	6
	Kent	0·74	6
	Dorman	0·68	6
	L.S.D	0·04	
2		(cm s^{-1})	(number of days)
Mean: T=28° C	Clark	1·0	3
VPD=2·3 kPa	Kent	1·0	3
	Dorman	1·0	3
Mean: T=34° C	Clark	1·0	5
VPD=3·4 kPa	Kent	0·83	5
	Dorman	0·71	5
	L.S.D	0·08	

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Table 3. *Ratios of root weight to leaf area, leaf conductance (c_s) to water vapour at 3·0 compared to 1·0 kPa leaf to air water vapour pressure difference, and conductance to water for three cultivars of* Glycine max *and two lines from cv. Kent. Plant conductance was estimated as transpiration rate divided by the difference between substrate and leaf water potentials for plants at 30° C air temperature, 10° C dewpoint, and expressed per unit of leaf area. Root conductances were estimated from water flow through root systems pressurised to 1·0 MPa.*

Cultivar	Root wt/area ($g\ m^{-2}$)	c_s at 3·0/ c_s at 1·0	Plant conductance ($mmol\ s^{-1}\ m^{-2}\ MPa^{-1}$)	Root conductance ($mmol\ s^{-1}\ m^{-2}\ MPa^{-1}$)
Clark	26	0·80	12·0	13·9
Kent A	22	0·72	10·1	11·4
Kent	21	0·65	9·3	10·8
Kent B	19	0·62	8·5	9·3
Dorman	18	0·56	7·8	8·8
L.S.D.	2	0·06	1·3	1·0

potentials were equal for the cultivars. In order for leaf water potentials to be equal, leaf conductance at a given VPD must be a function of plant conductance. The observation (Maier-Maercker, 1981) that stomatal aperture at high VPD was directly related to the water potential of the root medium, observations by several workers of increased sensitivity of leaf conductance to VPD in dry soil (Ludlow, 1980) and the observed decrease in leaf conductance at constant leaf water potential with soil drying (Bates & Hall, 1981) could all be consistent with a dependence of the sensitivity of leaf conductance to VPD on soil-plant conductance. This idea is not contradictory to the data showing high resistance to water flow between the epidermis and mesophyll (Sheriff, 1982), because there are no data about what fraction of its water supply the epidermis receives from the xylem via mesophyll cells.

Partial closure of stomata at high VPD has both potential advantages (e.g. reduction of water loss) and disadvantages (reduction in net photosynthesis). When conservation of soil water is likely to be of substantial benefit to a crop, then a decrease in leaf conductance in response to high VPD may improve water use efficiency and yield (Cowan, 1977; Cowan & Farquhar, 1977). If soil water is unlikely to be limiting, then lack of response of leaf conductance to VPD could improve photosynthetic production, provided that midday leaf water deficits were not injurious. We know little about the importance of midday water deficits to long-term growth (Kramer, 1983). There is not yet sufficient understanding to predict whether growth or yield would be improved by increasing or by decreasing the sensitivity of stomata to VPD in humid climates (Ludlow, 1980). Comparison of lines differing in sensitivity should aid in achieving this understanding.

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